

Dexter Fernandes

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Computer Vision Engineer

Computer Vision Engineer building real-time video analytics, production IoT and edge perception systems for British Petroleum and Austin Industries. Experienced with NVIDIA DeepStream 6.x/9.0, GStreamer, Python and C++17 across multi-camera ingestion, inference, tracking, monitoring and field debugging under latency, compute and memory constraints. I thrive in a dynamic startup environment with an interest in contributing to sustainability and transport.

PROJECTS

DeepStream RTSP Pipeline

Jun 2026

- Built an end-to-end **NVIDIA DeepStream 9.0/GStreamer RTSP video analytics pipeline** for multi-stream computer vision, using rtspsrc, nvstreammux, nvinfer, nvtracker, and nvdsosd to run YOLO+NvDCF tracking, metadata extraction, anonymisation, CSV logging, and re-streamed output across 3 concurrent camera feeds.
- Deployed and optimised a **YOLO26n TensorRT FP16 detector** on **6GB VRAM Turing hardware**, combining dynamic-batch engine conversion, nvinfer integration, p99 latency profiling, and FP16-vs-FP32 accuracy validation to sustain **29.7 FPS per stream live** and **131.3 FPS per stream unthrottled**.
- Implemented a **C++/CUDA TensorRT decode plugin** using IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked tracker configurations across **IOU, NvDCF, and** ByteTrack-inspired **NvSORT**, producing MOTA, IDF1, ID switches, fragmentation, per-frame detection counts, FPS, and VRAM impact to guide production tracker selection.
- Added production hardening through **GPU smoke tests, model-promotion gates, structured JSON logging, and per-camera health metrics** for reactive debugging across liveness, dropped frames, reconnect events, and RSS.

WORK EXPERIENCE

GrowthStage • Remote

May 2026 - Present

Recruiter

- Evaluated candidates' technical depth across perception, SLAM, and deployment using hands-on robotics and computer vision knowledge to separate genuine signal from surface-level claims.

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed and maintained NVIDIA DeepStream 6.x video inference pipelines for 24/7 industrial perception deployments, supporting camera ingestion, model inference, and field diagnostics in harsh desert environments.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts reducing live video pipeline crashes by 67% and improved root-cause analysis.
- Led field validation on BP oil-rig deployments, diagnosing model failures under lighting, motion and environmental stressors and translating findings into practical fixes for live operation.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.
- Developed an internal web dashboard in Django to surface inference outputs, system health, and performance metrics, enabling faster debugging and root-cause analysis across deployed systems.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.

Lead Software Engineer - AI

- Led a team to design, train, and deploy vision pipelines for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries.
- Built low-latency asynchronous perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel hardware.
- Profiled and optimised C++ inference and pre and post-processing hotspots using Intel VTune Profiler for targeted performance fixes in our CNN inference pipeline.
- Ran systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.
- Mentored junior engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.

EDUCATION**Master of Science in Robotics - Merit**

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, VLP-16 and Intel RealSense D435i.
- Performed a JetPack platform upgrade on Jetson Xavier AGX to access newer CUDA and TensorRT, porting a ROS2/PCL SLAM stack across the upgrade and resolving the dependency and driver breakage it introduced.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **Edge AI & Video Pipelines:** NVIDIA DeepStream 6.x/9.0, GStreamer, RTSP video pipelines, nvstreammux, nvinfer, nvtracker, nvdsosd, CUDA, JetPack, camera behaviour debugging, optical performance eval, ISP tuning
- **Edge Inference and Optimisation:** FP16/INT8 deployment, OpenVINO, TensorRT, ONNX, custom TensorRT plugins, dynamic batching, latency/throughput profiling
- **Computer Vision & Image Processing:** Object detection, Object tracking, RT-DETR, YOLOv8, YOLO26, SSD MobileNet V2, EfficientDet, NvDCF, ByteTrack
- **Cloud, Edge & MLOps:** Apache Kafka, MLflow, Docker, CI/CD, Production monitoring, Structured logging, Health checks, model-promotion gates, failure-mode analysis
- **Programming & Tooling:** Python, C++17, Bash, Django, CMake, Git, Intel VTune, OpenCV, CUDA C++
- **Edge AI Hardware:** NVIDIA Jetson AGX Xavier, Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU